

William Dong

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Dedicated to providing autonomous ground solutions for defense technology. Mechanical & materials engineer, robotics-nonprofit founder, and two-time state-champion boxer.

EDUCATION

Carnegie Mellon University

Expected May 2030

B.S. Materials Science & Engineering, Incoming Freshman

Great Valley High School

2022 – 2026

Senior · Malvern, PA · GPA 4.0 / 4.0

Relevant coursework: AP Physics C (Mechanics, E&M), AP Calculus BC, AP CSA

EXPERIENCE

Founder & CEO

2024 – Present

Formula Theory & Automotive Design (FTAD) · 501(c)(3) Nonprofit

- Founded an F1-inspired robotics competition nonprofit, securing \$65,000+ in funding through targeted sponsorships and grant acquisition.
- Conducted extensive market research and obtained international product validation, establishing FTAD's competition framework across multiple countries.
- Engineered competition vehicle designs in Onshape and Fusion 360, applying CFD simulation and topology optimization to maximize aerodynamic and structural performance.

MechE Intern, UGV Defense Company

Summer 2026

Raptor Defense Company

- Led SCORPION, an expendable payload vehicle, as lead and sole mechanical engineer, owning the full mechanical design from concept to working prototype.
- Designed the complete assembly in Onshape in 6 days, then manufactured and assembled it in 3.
- Delivered on spec: exceeded 50 mph (conservative estimate), used under 50% of the R&D budget, and passed the preliminary test matrix.
- Built a Portable Control Unit for the electronics team in SolidWorks.

Design Lead & CAD Specialist

2024 – 2026

VEX Robotics · World-Qualified Competitive Team

- Led all mechanical design and CAD work on a team that qualified for the VEX World Championship in the 2024 season.

LEADERSHIP & ACTIVITIES

Co-Founder

2022 – 2024

NextGen Business & Law Forum · 501(c)(3) Nonprofit

- Organized entrepreneurship competitions for 500+ students across the United States.
- Oversaw event logistics, curriculum development, and partner outreach.

Future Business Leaders of America (FBLA)

2025

Web Development · National Qualifier · 4th in Pennsylvania

- Learned web-dev logic in 4 months and built the competition project in a team of 3.

Competitive Boxer

2023 – Present

USA Boxing

- Two-time PA State Champion (2024 & 2025).
- Volunteer boxing coach, mentoring athletes ages 12–34 in technique, discipline, and sportsmanship.

SKILLS

CAD & Design: Onshape, Fusion 360, SolidWorks, CFD Simulation, Topology Optimization

Programming: Python, Java, C++

Engineering: Kinematic Systems, Structural Analysis, Aerodynamic Simulation, Rapid Prototyping, UGV Systems

INTERESTS

Kinematics engineering, automotive engineering & racing technology, unmanned systems, defense technology, entrepreneurship, and combat sports.